Holland Blumer

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EXPERIENCE

Recurse Center – *Participant* **Mar 2026 – Present**

- **Poster BluePrint** – An AI-assisted poster analysis tool that identifies design elements in uploaded posters that are recreatable with modern creative coding tools, using OpenCV/SIFT image matching with Google Gemini 2.5 Flash as a fallback, Next.js, TypeScript, FastAPI, GraphQL, Postgres, SQLAlchemy, Python, p5.js, SVG filters, Three.js, and GLSL shaders.
- **3D Motion Marbling** – An interactive digital experience that transforms live hand movements into a virtual marbling comb, displacing pixels on an HTML canvas using webcam input, MediaPipe hand tracking mapped to a MANO hand model overlay, OpenCV frame processing, JavaScript, Python, FastAPI, Google Cloud Storage (GCS), and Render.
- **Creative Coding NYC Gallery** – An online gallery developed as part of my role as an organizer of Creative Coding NYC, showcasing posters created by community members using Next.js, TypeScript, Three.js, WebGL, GLSL shaders, and a SQLite-backed REST API for rotating weekly features per community feedback.

Mercor and Independent Clients – *Contract Software Engineer* **Jun 2022 – Present**

- Lead end-to-end design and development for 12+ recurring clients across retail, real estate, restaurants, and private equity, translating client needs and brand identity into custom UX/UI through iterative wireframing and direct client feedback, then shipping and maintaining production sites including cherylfudge.com, meredithwnorvell.com, and katherinegroverfinejewelry.com.
- Run agentic AI coding workflows in containerized environments using OpenCode with open-source LLMs (GLM 4.7, Kimi K2-Thinking, MiniMax M2.5) to solve engineering tasks, evaluate generated solutions against test suites, and identify specification gaps, then write evaluation rubrics and review code patches to benchmark AI reasoning and implementation quality, using Docker, Git, Python, JavaScript, and C.
- **Divot** – A B2C web app for sustainability, designing the system end-to-end, using React, JavaScript, HTML, CSS, Adobe Illustrator, AWS (Amplify, EC2, Lambda, S3, DynamoDB, IAM), and GraphQL.

Hyundai Motor Group – *Product Intelligence Intern* **Jun 2021 – Sep 2021**

- Led creation of a 100+ page internal engineering guide for compact, midsize, and BEV pickup development, running bi-weekly meetings with engineering heads across 8 teams to manage progress.
- Traveled to HMG's Irvine, CA site to observe consumer interviews in test vehicles alongside a senior market research analyst, documenting feedback that informed EV feature planning.

LARQ – *Product Development Intern* **Jun 2020 – Sep 2020**

- Collected and analyzed capacitance sensor data from Arduino-based prototypes using Python, NumPy, Matplotlib, and Excel, identifying outliers and refining detection thresholds for a filter-lifespan tracking algorithm that shipped in the LARQ mobile app with 500+ ratings at 4.5 stars.
- Conducted GD&T on water bottle cap drawings to define dimensional requirements.

EDUCATION

Dartmouth College – *M.Eng. in Computer and Electrical Engineering* **Grad. 2022**

- **Helios** – An internal quality-control system for ChargePoint that captures inspection images and data from QA stations on the production line, built through iterative feedback from employees to define the charger detail view, key manufacturing metrics, and an employee notes feature, using Python, SQL, GraphQL, React, HTML/CSS, JavaScript, and AWS (EC2, Lambda, Amplify, S3, DynamoDB).
- Relevant Courses – Machine Learning & Statistical Data Analysis (Python), Introduction to Smartphone Programming (Kotlin), Digital Image Processing (MATLAB), and Computer Animation (Maya)

Northwestern University – *B.S. in Manufacturing and Design Engineering, minor in Data Science* **Grad. 2021**

- **UV Sense** – An optimized NFC antenna design for a L'Oréal-sponsored wearable UV sensor later commercialized as My Skin Track UV, with antenna connectivity tested and validated through the assembly of PCB prototypes using electroplating, laser cutting, and soldering.